

RIM-BP2 is required for rapid neurotransmitter release through regulation of Ca^{2+} channel clustering at hippocampal mossy fiber terminals

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Rinako Miyano, Hirokazu Sakamoto, Kenzo Hirose, Takeshi Sakaba 

Graduate School of Brain Science, Doshisha University, Kyotanabe-shi, Kyoto 610-0394, Japan • Department of Pharmacology, Graduate School of Medicine, The University of Tokyo, Bunkyo-ku, Tokyo 113-0033, Japan • International Research Center for Neurointelligence (WPI-IRCN), The University of Tokyo, Bunkyo-ku, Tokyo 113-0033, Japan

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Abstract

Synaptic vesicles dock and fuse at the presynaptic active zone (AZ), the specialized site for transmitter release. AZ proteins play multiple roles such as recruitment of Ca^{2+} channels as well as synaptic vesicle docking, priming and fusion. However, the precise role of each AZ protein type remains unknown. At phasic synapses having high release probability, Rab3-interacting molecule-binding protein2 (RIM-BP2) is known to affect Ca^{2+} channel localization at AZs. In order to dissect the role of RIM-BP2 at tonic synapses having low release probability, we applied electrophysiological recording and super-resolution imaging to hippocampal mossy fiber terminals of RIM-BP2 KO mice. By using whole cell capacitance measurements, we found that reduced Ca^{2+} currents were responsible for the decreased rates of transmitter release in RIM-BP2 KO terminals. Consistently, STED microscopy pictured lower densities of P/Q-type Ca^{2+} channels at AZs deficient in RIM-BP2. Our results suggest that the RIM-BP2 function differs between phasic and tonic synapses.

eLife assessment

Miyano et al. study the impact of RIM-BP2 deletion at mossy fiber synapses, using direct electrophysiological recordings from mossy terminals and STED super-resolution microscopy. The paper addresses an **important** question in the field of synaptic transmission and provides **compelling** evidence demonstrating reduced calcium channel abundance in mossy terminals upon RIM-BP2 removal.

Introduction

At chemical synapses, an action potential arriving at the presynaptic terminal opens voltage-gated Ca^{2+} channels, and Ca^{2+} influx through voltage-gated Ca^{2+} channels triggers neurotransmitter release from synaptic vesicles within ms. Synaptic vesicle fusion occurs at a specialized region of

the presynaptic membrane called the active zone (AZ), where Ca^{2+} channel clusters and vesicle fusion sites are accommodated via AZ scaffold proteins (Südhof, 2012). For rapid transmission, accumulation of Ca^{2+} channels, co-localization of Ca^{2+} channels and fusion sites, and molecular priming of synaptic vesicles for fusion are required, and the differences between these factors might provide a basis for synaptic diversity. However, it remains unknown how these factors are regulated by AZ-scaffold proteins.

Rab3-interacting molecule-binding proteins (RIM-BPs) represent one principal, conserved family of AZ proteins and bind to RIMs and Ca^{2+} channels (Wang et al., 2000; Hibino et al., 2002). RIM-BP family includes RIM-BP1, RIM-BP2, and RIM-BP3 in mammals, but RIM-BP3 expression is low in the nervous system (Mittelstaedt and Schoch, 2007). In *Drosophila* neuromuscular junctions (NMJs), loss of RIM-BPs decreases Ca^{2+} channel density and reduces release probability (Liu et al., 2011). Moreover, RIM-BP here is necessary for tight coupling of synaptic vesicles to Ca^{2+} channels and replenishment of high release probability vesicles at *Drosophila* NMJs (Liu et al., 2011; Müller et al., 2015; Petzoldt et al., 2020). In mammalian synapses, the observations from KO mice are diverse. In RIM-BP1,2 DKO mice, the coupling between Ca^{2+} channels and synaptic vesicles became loose, and action potential-evoked neurotransmitter release was reduced at the calyx of Held synapse (Acuna et al., 2015). At hippocampal CA3-CA1 synapses, RIM-BP2 deletion alters Ca^{2+} channel localization at the AZs without altering total Ca^{2+} influx. Besides, RIM-BP1,2 DKO has no additional effect, indicating that RIM-BP2 dominates the function of RIM-BP isoforms (Grauel et al., 2016). Tight coupling between Ca^{2+} channels and vesicles is crucial for rapid and efficient transmitter release (Wadel et al., 2007).

In contrast, hippocampal mossy fiber-CA3 synapses are characterized by low release probability (Vyleta and Jonas, 2014). At hippocampal mossy fiber synapses, gSTED imaging suggested that RIM-BP2 KO altered the Munc13-1 cluster number and distribution. In addition, RIM-BP2 deletion decreased the number of docked vesicles (Brockmann et al., 2019). However, because of technical difficulty, direct and quantitative measurements of exocytosis in RIM-BP2 KO terminals have not been performed so far.

To quantify the RIM-BP2 function in mossy fiber-CA3 synapses, we performed whole-cell patch-clamp recordings from WT and RIM-BP2 KO hippocampal mossy fiber boutons. We measured the size of the readily releasable pool (RRP) of synaptic vesicles, Ca^{2+} influx, Ca^{2+} sensitivity of vesicle fusion, and the sensitivity of release to an intracellular Ca^{2+} buffer. From electrophysiological recordings, we show that Ca^{2+} current amplitudes are reduced in RIM-BP2 KO terminals, and that the reduction of Ca^{2+} currents is the dominant factor for that of transmitter release. Using STED imaging, we find the densities of P/Q-type Ca^{2+} channels in terminals to be reduced in RIM-BP2 KO. We suggest that RIM-BP2 regulates the number of P/Q-type Ca^{2+} channels at the AZ and is critical for rapid transmitter release at hippocampal mossy fiber terminals.

Results

RIM-BP2 KO reduces synaptic vesicle release and calcium currents at hippocampal mossy fiber boutons

To investigate the roles of RIM-BP2 in synaptic transmission, we examined the kinetics of exocytosis and Ca^{2+} influx in WT and RIM-BP2 KO synapses. We applied whole-cell patch-clamp recordings and capacitance measurements to hippocampal mossy fiber terminals in acute slices from WT and RIM-BP2 KO mice, in order to measure Ca^{2+} currents and exocytosis directly. The patch pipette contained Cs^+ -based internal solution and 0.5 mM EGTA to isolate Ca^{2+} currents (Midorikawa and Sakaba, 2017). The external solution contained 1 μM TTX to block Na^+ currents. Membrane capacitance was measured by applying a sinusoidal wave (1000 Hz, ± 30 mV in amplitudes, Lindau and Neher, 1988). When the terminal was depolarized from -80 mV to $+10$

mV for 10 ms, Ca^{2+} currents and membrane capacitance were recorded as shown in **Fig. 1A**. During the depolarizing pulse, capacitance was not measured due to large conductance changes, and the increase was used for measuring synaptic vesicle exocytosis. To determine the size of the RRP and the time course of exocytosis, we measured capacitance changes (ΔC_m) in response to various durations of depolarization. Here, the length of the depolarizing pulse was varied between 5 ms and 100 ms, and ΔC_m plotted against pulse duration (**Fig. 1B**, bottom). ΔC_m in response to a 10 ms pulse was reduced in RIM-BP2 KO ($p = 0.0499$). ΔC_m became larger as the duration of pulse was prolonged, but the increase started to be saturated at a 100 ms pulse in WT terminals, suggesting depletion of the RRP. ΔC_m in response to a 100 ms pulse was 52 ± 9.1 fF ($n = 8$). The amplitude corresponds to the fusion of about 500-600 synaptic vesicles (Hallermann et al., 2003; Midorikawa and Sakaba, 2017). The time course of ΔC_m could be fitted by a single exponential with a time constant of 58 ± 11 ms. In RIM-BP2 KO terminals, ΔC_m in response to a 100 ms pulse was smaller (41 ± 5.9 fF, $n = 8$) than that of WT terminals, and the release time constant was slower (64 ± 14 ms) than in WT terminals. We should note that the time constant is not necessarily reliable because the ΔC_m values do not reach a plateau value at 100 ms pulse in some terminals. The amplitudes of Ca^{2+} currents were smaller in RIM-BP2 KO terminals as compared to WT terminals (**Fig. 1B**, top). The Ca^{2+} current in response to a 100 ms pulse was reduced by ~30% in RIM-BP2 KO terminals (27 ± 3.9 pA, $n = 8$) compared to WT terminals (41 ± 4.3 pA, $n = 8$) ($p = 0.0379$). Thus, the results show that RIM-BP2 deletion reduces both neurotransmitter release and Ca^{2+} influx.

To investigate the Ca^{2+} influx in more detail, we examined the voltage dependence of Ca^{2+} currents. Ca^{2+} currents were recorded in the presence of $1 \mu\text{M}$ TTX, a Na^+ channel blocker, and 10 mM TEA, a K^+ channel blocker. **Fig. 1C** shows voltage step protocols and representative current traces. Ca^{2+} currents were elicited by 5 ms depolarizing pulses from a holding potential of -80 mV to various potentials to obtain I-V relationships. In both genotypes, Ca^{2+} currents started to be activated at around -20 mV, and maximal amplitude was observed at $\sim +10$ mV (**Fig. 1D**). There was no major difference in the voltage dependence of Ca^{2+} currents between WT and RIM-BP2 KO mice. Consistently, activation time constants of Ca^{2+} currents were similar between WT and RIM-BP2 KO, when the terminal was depolarized to $+10$ mV ($\tau = 1.4 \pm 0.2$ ms in WT, $n = 6$; $\tau = 1.3 \pm 0.2$ ms in KO, $n = 4$). These results suggest a reduction in the number of Ca^{2+} channels rather than changes in activation kinetics is responsible for the current decrease observed.

High extracellular calcium concentration rescues the decrease of synaptic vesicle release in RIM-BP2 KO

Ca^{2+} influx controls exocytosis by regulating vesicular release probability and/or the RRP size (Katz, 1969; Thanawala and Regehr, 2013). Because Ca^{2+} currents were reduced in RIM-BP2 KO mice, we tested whether the reduction of release in RIM-BP2 KO mice could be explained by that of Ca^{2+} influx or impaired synaptic vesicle docking / priming (Brockmann et al., 2017). To increase Ca^{2+} influx, we raised the extracellular Ca^{2+} concentration ($[\text{Ca}^{2+}]_{\text{ext}}$) from 2 mM to 4 mM. Ca^{2+} currents and membrane capacitances were recorded at 4 mM $[\text{Ca}^{2+}]_{\text{ext}}$, and amplitudes then plotted against pulse duration (**Fig. 2A**). Although Ca^{2+} current elicited by a 100 ms pulse in RIM-BP2 KO was still smaller (33 ± 2.7 pA, $n = 3$) than that in WT (47 ± 6.6 pA, $n = 4$) (**Fig. 2B**, top), elevated $[\text{Ca}^{2+}]_{\text{ext}}$ increased Ca^{2+} currents in both WT and RIM-BP2 KO mice. At 4 mM $[\text{Ca}^{2+}]_{\text{ext}}$, ΔC_m in response to a 100 ms pulse in RIM-BP2 KO was larger (51 ± 11 fF, $n = 3$) than at 2 mM $[\text{Ca}^{2+}]_{\text{ext}}$ and indeed showed a similar value in WT at 2 mM $[\text{Ca}^{2+}]_{\text{ext}}$ (**Fig. 2B**, bottom). At the same time, ΔC_m in WT was not altered (53 ± 3.2 fF, $n = 4$) when higher $[\text{Ca}^{2+}]_{\text{ext}}$, suggesting that the RRP in WT was already depleted at 2 mM $[\text{Ca}^{2+}]_{\text{ext}}$. Also, the time course of exocytosis at 4 mM $[\text{Ca}^{2+}]_{\text{ext}}$ was similar to that of WT at 2 mM $[\text{Ca}^{2+}]_{\text{ext}}$, as the time courses were superimposable in **Fig. 2B**. These results suggest that a decrease in neurotransmitter release in RIM-BP2 KO could be rescued by an increase in Ca^{2+} influx, easiest explained by a reduction of Ca^{2+} channel number.

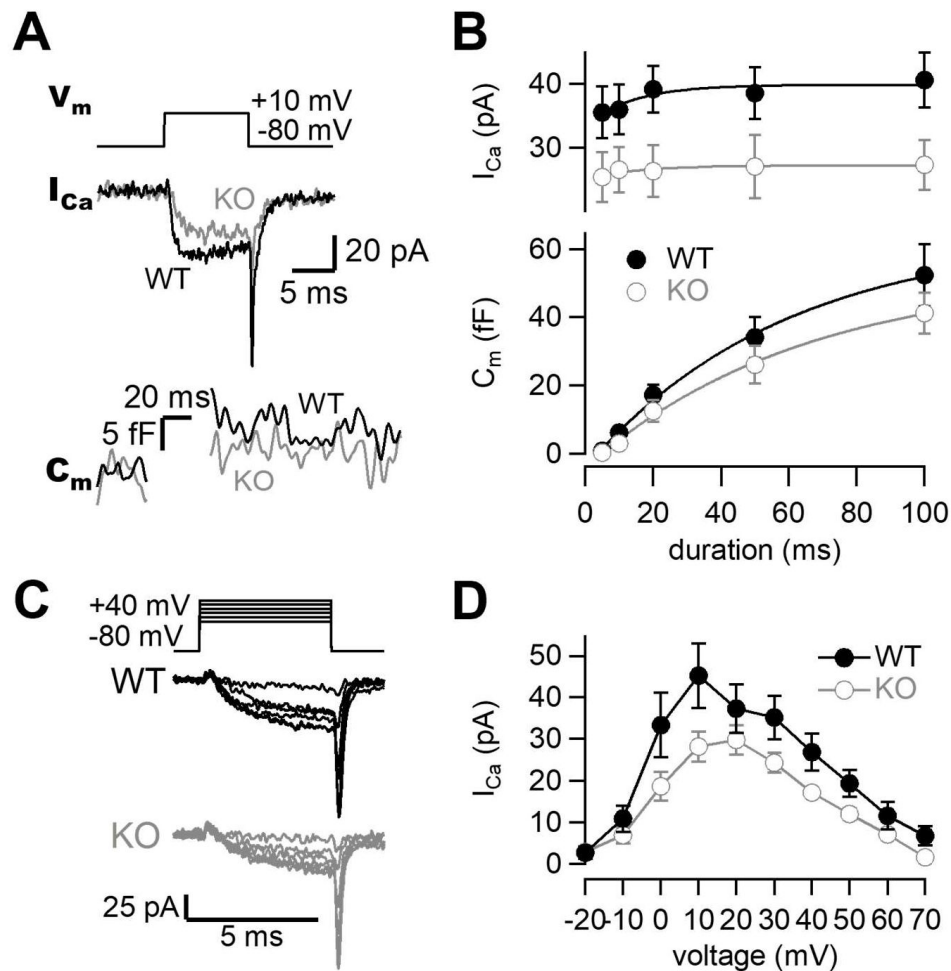


Figure 1.

RIM-BP2 KO reduces calcium currents and synaptic vesicle release

(A) The terminal was depolarized from -80 mV to $+10$ mV in WT (black) and RIM-BP2 KO (gray) hippocampal mossy fiber boutons. Ca^{2+} currents (I_{Ca}) and membrane capacitance (C_m) in response to a 10 ms pulse (V_m) are shown. (B) (top) The peak Ca^{2+} currents were plotted against the pulse duration. At 10 ms, $p = 0.0499$ (Wilcoxon rank sum test). (bottom) The capacitance jumps were plotted against the pulse duration. At 20 and 100 ms, $p = 0.0379$ and 0.0379 , respectively (Wilcoxon rank sum test). Black filled circles and gray open circles represent the data from WT ($n = 8$) and RIM-BP2 KO ($n = 8$), respectively. Each data point represents mean $\pm$ SEM. (C) Experimental protocol and representative traces for presynaptic Ca^{2+} current measurements in WT (black) and RIM-BP2 KO (gray) boutons. Terminals were sequentially depolarized for 5 ms with 2 ms intervals from -80 mV to $+70$ mV by 10 mV steps. (D) Current-voltage relationships of peak Ca^{2+} currents in WT (black filled circle; $n = 4-6$) and RIM-BP2 KO (gray open circle; $n = 4-5$). I_{Ca} s were elicited by 5 ms depolarizations. Each data point represents mean $\pm$ SEM. Numerical values of plots are provided in [Figure 1-source data 1](#).

Source data 1. Datasets of Ca^{2+} -current and capacitance amplitudes presented in [Figure 1](#).

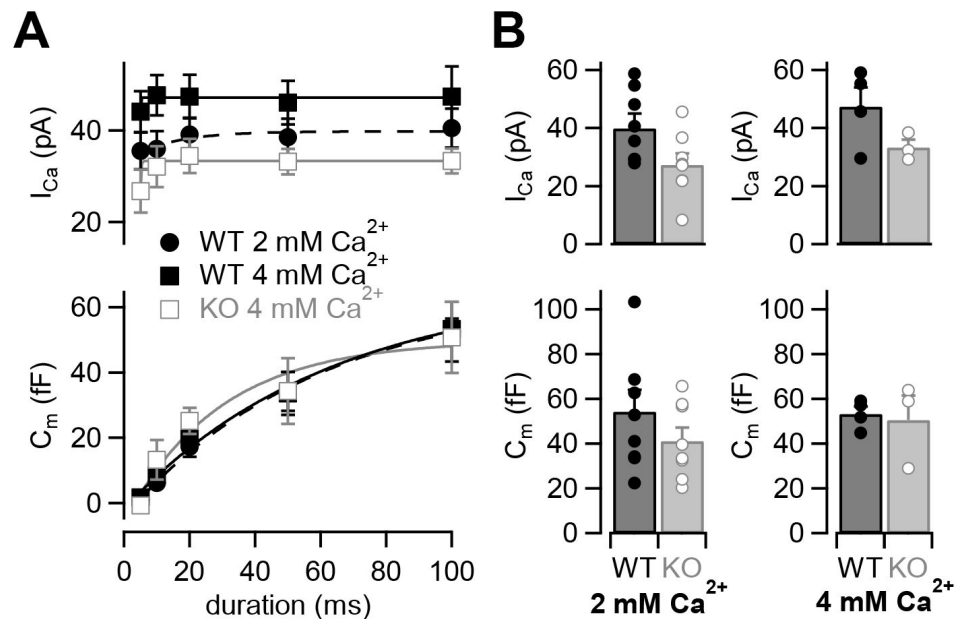


Figure 2.

High extracellular calcium concentration rescues the decrease of synaptic vesicle release, but does not affect the RRP size

(A) (top) The I_{Ca} s were plotted against the pulse duration in 4 mM $[Ca^{2+}]_{ext}$. (bottom) The C_m s were plotted against the pulse duration in 4 mM $[Ca^{2+}]_{ext}$. Black filled squares and gray open squares represent the data from WT ($n = 4-6$) and RIM-BP2 KO ($n = 3-4$), respectively. For comparison, the WT data in 2 mM $[Ca^{2+}]_{ext}$ are superimposed (black filled circle; $n = 8$) (the same data set as Fig. 1B). Each data point represents mean $\pm$ SEM. (B) Average I_{Ca} s (top) and C_m s (bottom) in response to a 100 ms pulse in WT (black bars) and RIM-BP2 KO (gray bars) terminals. Extracellular Ca^{2+} concentration was 2 mM (left) or 4 mM (right). Error bars show SEM. Circles indicate individual values. Numerical values of plots are provided in Figure 2-source data 1.

Source data 1. Datasets of Ca^{2+} -current and capacitance amplitudes presented in Figure 2.

Decreased Calcium currents are mainly responsible for reduced release in RIM-BP2 KO

In order to further address whether the reduction of presynaptic Ca^{2+} influx observed was responsible for the compromised release of RIM-BP2 KO mice, we adjusted Ca^{2+} current amplitudes of WT to that of RIM-BP2 KO by lowering $[\text{Ca}^{2+}]_{\text{ext}}$. The amplitudes of Ca^{2+} currents and capacitance jumps were plotted against pulse duration (**Fig. 3A**). At 1 mM $[\text{Ca}^{2+}]_{\text{ext}}$, Ca^{2+} currents and ΔC_m were smaller than those of RIM-BP2 KO at 2 mM $[\text{Ca}^{2+}]_{\text{ext}}$. At 1.5 mM $[\text{Ca}^{2+}]_{\text{ext}}$, Ca^{2+} currents became comparable to those of RIM-BP2 KO at 2 mM $[\text{Ca}^{2+}]_{\text{ext}}$. Here, the ΔC_m evoked by a 100 ms pulse (43 ± 6.7 fF, $n = 5$) was comparable between WT and RIM-BP2 KO (**Fig 3A**). The average time course of capacitance increase of WT at 1.5 mM $[\text{Ca}^{2+}]_{\text{ext}}$ was similar to that of RIM-BP2 KO at 2 mM $[\text{Ca}^{2+}]_{\text{ext}}$. Therefore, by reducing Ca^{2+} currents, WT data could reproduce the release time course of RIM-BP2 KO. In **Fig. 3B**, ΔC_m in response to a 100 ms pulse at various $[\text{Ca}^{2+}]_{\text{ext}}$ s were plotted against the peak Ca^{2+} current amplitudes. The relationship could be fitted by a Hill plot with $n = 3$ in WT (Schneggenburger et al., 1999). Consistently, when capacitance jumps elicited by various pulse durations are plotted against total Ca^{2+} influx (Fig. S1), the RIM-BP2 KO data were superimposed on the WT data, indicating that Ca^{2+} -sensitivity for release was not altered and that the reduction of Ca^{2+} currents determine the reduced release in RIM-BP2 KO. In **Fig. 3C**, relative amplitudes of Ca^{2+} currents and capacitance in response to a 100 ms pulse are plotted. The RIM-BP2 KO data were superimposed on the WT data. These results confirmed that synaptic vesicles have similar sensitivity to Ca^{2+} influx in both genotypes.

High EGTA experiments suggest unaltered coupling between calcium channels and synaptic vesicles

Previous studies have used Ca^{2+} chelator such as EGTA to examine the sensitivity of release to intracellular Ca^{2+} buffers (Adler et al., 1991; Borst & Sakmann, 1996; Vyleta & Jonas, 2014). When the physical coupling between Ca^{2+} channels and synaptic vesicles is tight, EGTA (slow Ca^{2+} chelator) has less effect on release probability, but EGTA is effective when the coupling is loose. Previous experiments have suggested that RIM-BPs regulate the channel-vesicle coupling at the calyx of Held (Acuna et al., 2015). If this observation also applies to the mossy fiber synapse, EGTA sensitivity of release should be higher in RIM-BP2 KO terminals. To investigate the coupling at mossy fiber boutons in RIM-BP2 KO, we changed the concentration of EGTA in the patch pipette. In the presence of 5 mM EGTA, we depolarized the terminal from -80 mV to $+10$ mV for 5 ms and recorded Ca^{2+} currents and membrane capacitance (**Fig. 4A**). Ca^{2+} currents and ΔC_m were plotted against pulse duration (**Fig. 4B**). In both WT and RIM-BP2 KO mice, ΔC_m were reduced under 5 mM EGTA compared with the control condition of 0.5 mM EGTA. We compared the effect of EGTA on average amplitudes of peak Ca^{2+} currents in response to a 20 ms pulse (**Fig. 4D**). Interestingly, Ca^{2+} current amplitudes were ~ 1.5 times larger at 5 mM EGTA (53 ± 12 pA, $n = 5$) than at 0.5 mM EGTA (39 ± 3.7 pA, $n = 8$) in WT, though the data scattered under 5 mM EGTA condition and there was no statistically significant difference ($p = 0.524$). In RIM-BP2 KO, we did not observe such an increase. EGTA might inhibit Ca^{2+} channel inactivation in WT (von Gersdorff and Matthews, 1996). In **Fig. 4C**, we compared the effect of EGTA on the ΔC_m in response to a 20 ms pulse. In both genotypes, ΔC_m values were inhibited by 5 mM EGTA, and the inhibition was larger in RIM-BP2 KO.

Ca^{2+} current amplitudes in RIM-B2 KO are smaller than in WT (**Fig. 4B**). It is possible that strong effect of high EGTA on release maybe due to reduced Ca^{2+} currents in RIM-BP2 KO. Therefore, we changed $[\text{Ca}^{2+}]_{\text{ext}}$ from 2 mM to 1 mM and adjusted Ca^{2+} current amplitudes of WT to the RIM-BP2 KO level. The capacitance increase and the amplitudes of Ca^{2+} currents became similar to those of RIM-BP2 KO (**Fig. 4C** and **Fig. 4D**). These results indicate that WT and RIM-BP2 KO terminals have similar sensitivity of neurotransmitter release to EGTA. Thus, it is unlikely that changes in the coupling distance are responsible for the reduced exocytosis of the mutants.

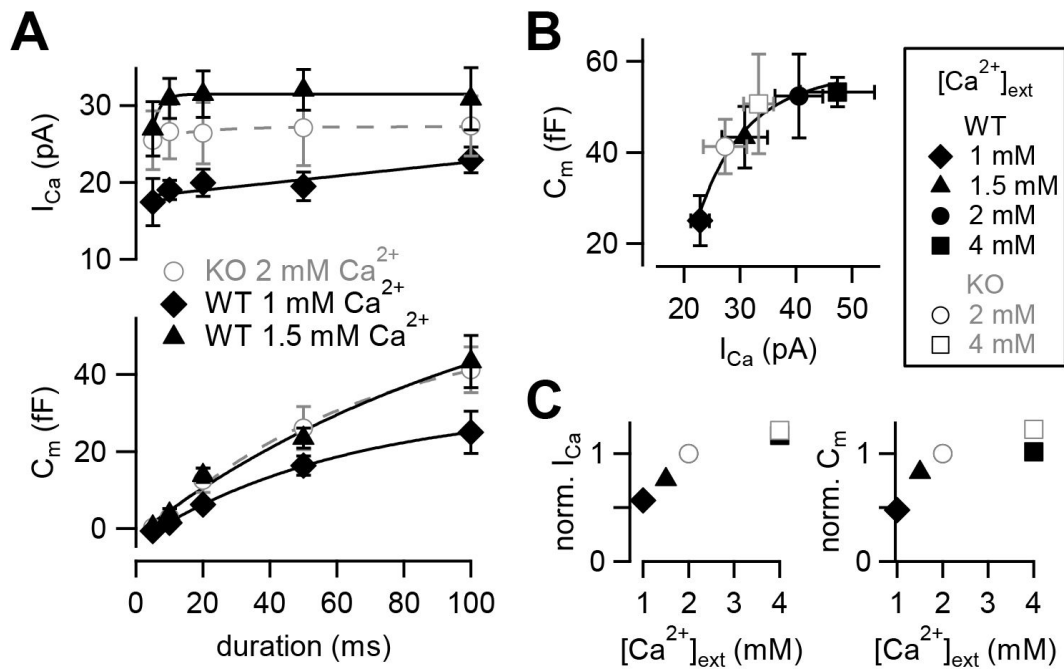


Figure 3.

Calcium-dependence of the release kinetics and the RRP size

(A) (top) The relationship between peak Ca^{2+} currents and pulse durations at different $[\text{Ca}^{2+}]_{\text{ext}}$ s. (bottom) The relationship between capacitance jumps and pulse durations at different $[\text{Ca}^{2+}]_{\text{ext}}$ s. Gray open circles, black diamonds and black triangles represent the data from RIM-BP2 KO at 2 mM $[\text{Ca}^{2+}]_{\text{ext}}$ ($n = 8$) (the same data set as [Fig. 1B](#)), WT at 1 mM $[\text{Ca}^{2+}]_{\text{ext}}$ ($n = 3-5$) and WT at 1.5 mM $[\text{Ca}^{2+}]_{\text{ext}}$ ($n = 5-7$), respectively. Each data point represents mean $\pm$ SEM. (B) Capacitance jumps at various $[\text{Ca}^{2+}]_{\text{ext}}$ s are plotted against calcium current amplitudes. Pulses were 100 ms depolarization from -80 mV to $+10$ mV. Each data point represents mean $\pm$ SEM. Data points were fitted with a Hill equation with $n = 3$. (C) (left) Average I_{Ca} s at indicated $[\text{Ca}^{2+}]_{\text{ext}}$ s were normalized to the response at 2 mM $[\text{Ca}^{2+}]_{\text{ext}}$ in each genotype. (right) Average C_m s at indicated $[\text{Ca}^{2+}]_{\text{ext}}$ s were normalized to the 2 mM $[\text{Ca}^{2+}]_{\text{ext}}$ response in each genotype. Numerical values of plots are provided in [Figure 3-source data 1](#).

Source data 1. Datasets of numerical values presented in [Figure 3](#).

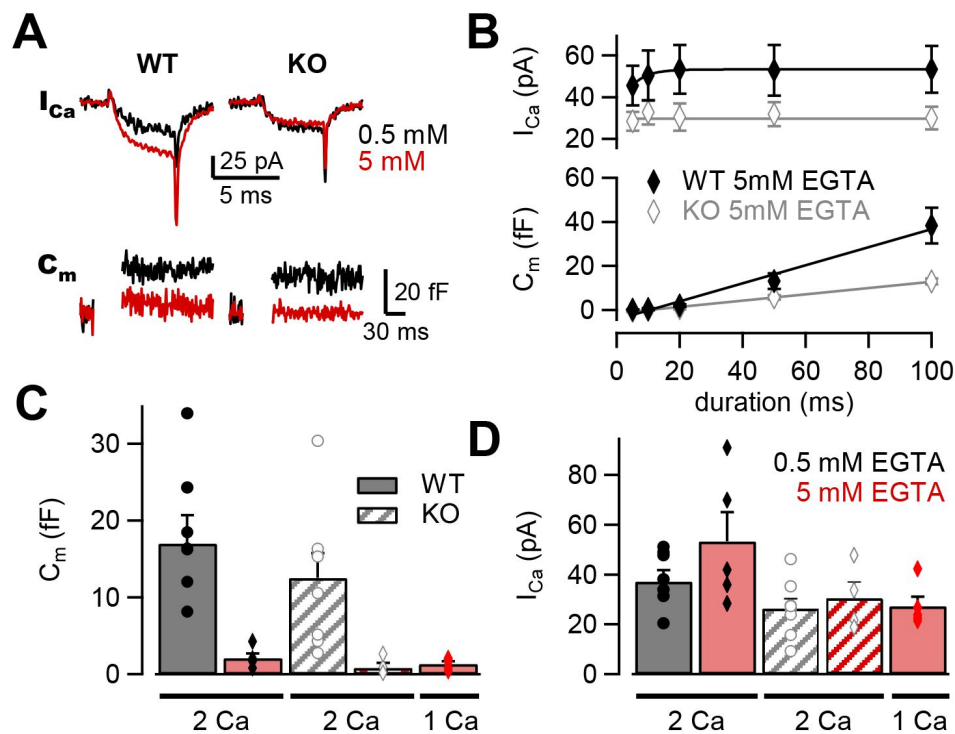


Figure 4.

The effects of high EGTA on calcium currents and synaptic vesicle exocytosis

(A) Example traces in response to a 5 ms depolarizing pulse to +10 mV in WT (left) and RIM-BP2 KO (right) boutons. Ca^{2+} current (I_{Ca}) and membrane capacitance (C_m) recorded with 0.5 mM EGTA (black) or 5 mM EGTA (red) in the patch pipette are shown. (B) The I_{Ca} s (top) and C_m s (bottom) are plotted against the pulse duration. The patch pipette contained 5 mM EGTA. Black filled diamonds and gray open diamonds represent the data from WT ($n = 4-5$) and RIM-BP2 KO ($n = 4$), respectively. Each data point indicates mean $\pm$ SEM. (C, D) Average C_m s (C) and I_{Ca} s (D) elicited by a 20 ms pulse in WT (filled bars) and RIM-BP2 KO (hatched bars). The extracellular Ca^{2+} concentration was 1 mM or 2 mM. The intracellular solution contained either 0.5 mM EGTA (black) or 5 mM EGTA (red). Error bars show SEM. Circles and diamonds indicate individual values. Numerical values of plots are provided in **Figure 4-source data 1**.

Source data 1. Datasets of Ca^{2+} -current and capacitance amplitudes presented in **Figure 4**.

STED microscopy reveals that RIM-BP2 KO reduces the density of P/Q-type Ca^{2+} channels

Our electrophysiological data suggest that RIM-BP2 may regulate the density of Ca^{2+} channels, or possibly Ca^{2+} channel properties. In order to directly study the Ca^{2+} channel density at the AZ, we performed STED microscopy analysis of Ca^{2+} channels in WT and RIM-BP2 KO hippocampal mossy fiber terminals. We here focused on P/Q-type and N-type Ca^{2+} channels because both Ca^{2+} channel types are relevant for transmitter release at this synapse (Castillo et al., 1994; Pelkey et al., 2006; Li et al., 2007). We performed immunohistochemistry on thin brain cryosections from WT and RIM-BP2 KO mice. STED microscopy confirmed complete loss of RIM-BP2 proteins in KO terminals (Fig. S2). We then immunohistochemically labeled the α -subunit of either P/Q-type Ca^{2+} channel (Cav2.1) or N-type Ca^{2+} channel (Cav2.2) and Munc13-1 (AZ/release site marker, Böhme et al., 2016; Sakamoto et al., 2018) with VGLUT1 (presynaptic marker) and PSD-95 (postsynaptic marker) (Fig. 5A, Fig. 5B and Fig. S3). In the stratum lucidum of the hippocampal CA3 region, mossy fibers form large synapses containing multiple and clustered AZs on CA3 pyramidal cells, and also small synapses containing a single AZ on interneurons (Acsády et al., 1998; Rollenhagen et al., 2007). Thus, we identified AZs using Munc13-1 STED images and quantified the signal intensity of Cav2.1 or Cav2.2 only at AZs clustered in large VGLUT1 positive terminals in the CA3 stratum lucidum, allowing us to confine the analysis to large mossy fiber terminals innervating onto CA3 pyramidal cells. The total signal intensity of Cav2.1 at the AZ was 22% lower in RIM-BP2 KO mice ($n = 4$ animals) than in WT mice ($n = 4$ animals) ($p = 0.0286$) (Fig. 5C, left). These data are consistent with the reduction of Ca^{2+} currents in RIM-BP2 KO mice. In contrast to Cav2.1, the total signal intensity of Cav2.2 and Munc13-1 did not significantly differ between WT and RIM-BP2 KOs ($p = 0.3143$ and $p = 0.0831$) (Fig. 5C, right). Furthermore, we found that the number of Cav2.1 clusters within the AZ identified by STED deconvolution analysis was not altered (2.5 ± 0.11 in WT, $n = 4$ animals; 2.3 ± 0.02 in KO, $n = 4$ animals) ($p = 0.114$). These data suggest that RIM-BP2 KO reduces the number of P/Q-type Ca^{2+} channels per cluster. To optically estimate physical distances between Ca^{2+} channels and release sites, we next analyzed the nearest neighboring distance between Cav2.1 and Munc13-1 clusters. These distances were unchanged (61 ± 1.4 nm in WT, $n = 4$ animals; 64 ± 1.3 nm in KO, $n = 4$ animals) ($p = 0.200$) (Fig. 5D), consistent with the similar sensitivity of release to EGTA between WT and RIM-BP2 KO mice. From these results, we conclude that decreased Ca^{2+} influx in RIM-BP2 KO hippocampal mossy fiber terminals is caused by reduction in the density of P/Q-type Ca^{2+} channels at the AZ.

Discussion

Accumulation of Ca^{2+} channels at the AZ, and efficient vesicle docking and priming are essential factors for fast synaptic vesicle exocytosis. AZ proteins are critical for fast synchronous release and synaptic diversity. RIM-BPs have been implicated in recruitment of Ca^{2+} channels to the AZ (Kaesler et al., 2011; Liu et al., 2011) by interacting with RIMs and Ca^{2+} channels (Wang et al., 2000; Hibino et al., 2002). In addition, RIM-BPs, like RIM1/2, recruit Munc-13 and accelerate synaptic vesicle priming (Brockmann et al., 2020). Concerning the relative importance of RIM-BPs on priming vs Ca^{2+} channel recruitment, we still do know little.

At murine central synapses, RIM-BPs deficient synapses demonstrate overall a rather mild decrease in the amounts of transmitter release (Acuna et al., 2015; Grauel et al., 2016; Luo et al., 2017; Krinner et al., 2017; Brockmann et al., 2019; Butola et al., 2021). Most studies so far have been focused on synapses with high release probability (phasic synapses) such as the calyx of Held and CA3-CA1 synapses. This said, at mossy fiber-CA3 synapses, KO of RIM-BPs has a strong influence on synaptic transmission (Brockmann et al., 2019). Notably, at mossy fiber-CA3 synapses, release probability is relatively low (tonic synapses). It was unknown whether the reduction of transmitter release at RIM-BP2 KO mossy fiber synapses is due to impaired Ca^{2+}

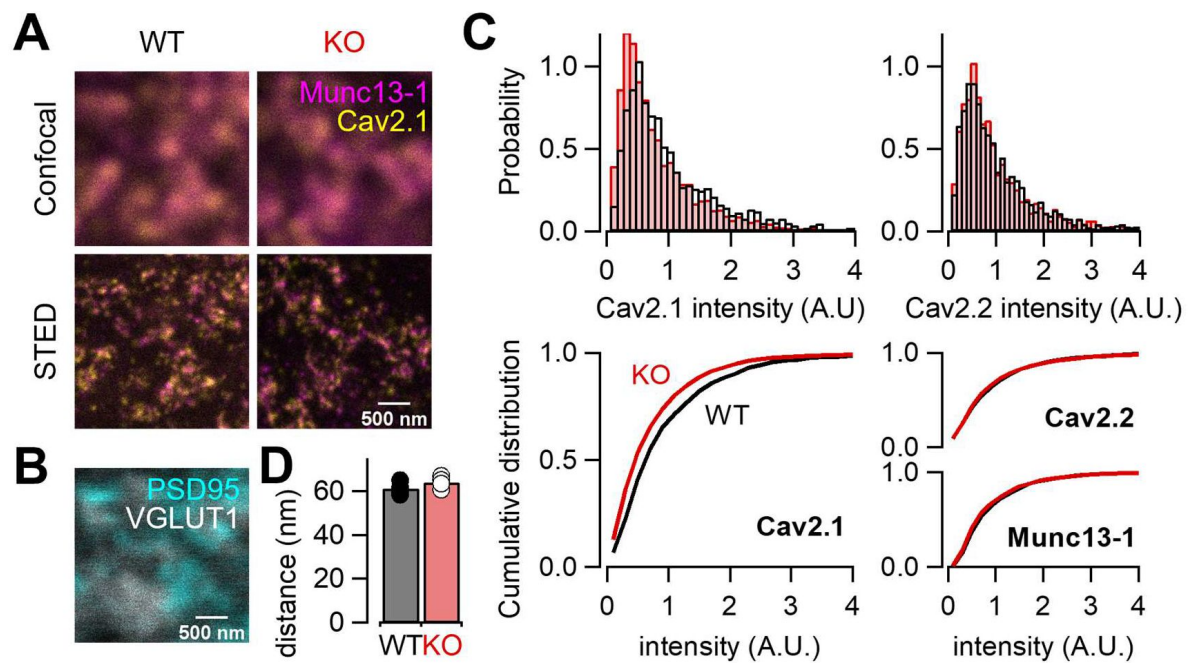


Figure 5.

RIM-BP2 deletion alters the density of Cav2.1 clusters within the AZ

(**A**) Confocal (*top*) and STED (*bottom*) images of Munc13-1 (magenta) and Cav2.1 (yellow) clusters at hippocampal mossy fiber terminals of WT (*left*) and RIM-BP2 KO (*right*) mice. (Scale bar: 500 nm.) (**B**) Confocal images of PSD95 (cyan) and VGLUT1 (white) to identify glutamatergic synapses in CA3 stratum lucidum: mossy fiber-CA3 synapses. (Scale bar: 500 nm.) The image is taken from same region shown in (**A**), (*left*). (**C**) Histograms (*top*) and cumulative distribution plots (*bottom*) of the total signal intensity of Cav2.1 (*left*), Cav2.2, and Munc13-1 (*right*) at AZs in WT (black) and RIM-BP2 KO (red) mice. (**D**) The average nearest neighbor distance between Cav2.1 and Munc13-1 clusters in WT (black; $n = 4$ animals) and RIM-BP2 KO (red; $n = 4$ animals) mice. Several hundreds of AZs per image were analyzed. Data show the average value of distance per animal and error bars represent SEM. Each data point indicates individual values. Numerical values of plots are provided in **Figure 5-source data 1**.

Source data 1. Datasets of numerical values presented in **Figure 5**.

channel recruitment, compromised vesicle docking or priming. Using STED microscopy, Brockmann et al. (2019) [\[1\]](#) suggested that the physical recruitment of Munc13-1 might be an important function of RIM-BP2, though functional demonstration had been lacking.

By using direct presynaptic patch clamp recordings, we here observed a decrease of Ca^{2+} current amplitudes ($\sim 30\%$) in RIM-BP2 KO mice (**Fig. 1** [\[1\]](#)). As the rates of exocytosis steeply depend on Ca^{2+} influx (Dodge and Rahamimoff, 1967 [\[2\]](#)), the reduced Ca^{2+} influx is likely sufficient to explain the slowed release time course at RIM-BP2 KO mossy fiber synapses. Moreover, STED microscopy supported decreased density of P/Q-type Ca^{2+} channels (Cav2.1) in the mutant mossy fiber terminal. Brockmann et al. (2019) [\[1\]](#) observed a reduction of Munc13-1 cluster number and docked vesicles in RIM-BP2 deficient synapses at hippocampal mossy fiber terminals. They identified the terminals by using ZnT3, enriched at mossy fiber (Brockmann et al., 2019 [\[1\]](#); Wenzel et al., 1997 [\[3\]](#)). Our STED imaging surprisingly did not detect differences in Munc13-1 cluster number at the AZ between WT and RIM-BP2 KO mice. Notably, we analyzed the number of Cav2.1, Cav2.2 and Munc13-1 clusters only within AZs showing direct co-localization with VGLUT1 and PSD-95, which mainly reflects the AZs facing CA3 pyramidal cells rather than the synapses onto interneurons (Ácsády et al., 1998 [\[4\]](#); Røllenhagen et al., 2007 [\[5\]](#)). Such a difference in areas analyzed between previous and our study might explain the difference. Still, our results do not exclude roles of RIM-BP2 in synaptic vesicle priming. We however suggest that the reduced Ca^{2+} influx might be more critical here when compared to effects of impaired vesicle priming.

Recent studies have reported that RIM-BPs deletion altered Ca^{2+} channel localization and loosened the coupling between Ca^{2+} channels and synaptic vesicles at phasic synapses (Acuna et al., 2015 [\[6\]](#); Grauel et al., 2016 [\[7\]](#); Butola et al., 2021 [\[8\]](#)). We investigated the Ca^{2+} channel-vesicle coupling at hippocampal mossy fiber terminals. The sensitivity of release to Ca^{2+} chelator EGTA was not changed between WT and RIM-BP2 KO mice. We also indicated RIM-BP2 KO does not alter the coupling of Ca^{2+} channels to release sites by using STED microscopy.

At phasic synapses, deletion of RIM-BP2 affects physical distance between Ca^{2+} channels and synaptic vesicles. In contrast, at hippocampal mossy fiber terminals having low release probability (Vyleta and Jonas, 2014 [\[9\]](#)), RIM-BP2 KO decreased the number of P/Q-type Ca^{2+} channels, thereby reducing Ca^{2+} influx and neurotransmitter release. Thus, we hypothesize the specific molecular-architectural and biochemical contributions by RIM-BPs might be of different relevance between phasic and tonic synapses. Brockmann et al. (2020) [\[10\]](#) proposed that RIM-BP2 controls Ca^{2+} channel recruitment and vesicle priming by profound interaction with RIMs and Munc13s. These results have been obtained through studying hippocampal cultures dominated by phasic synapses. It seems possible that the interaction among three protein types is tight at phasic synapses, simply because of high density of these proteins at AZs. Alternatively, one might speculate about additional proteins, different interaction surfaces taken, transsynaptic columns (Tang et al., 2016 [\[11\]](#)). Deletion of RIM-BP2 does not lead to loss of Ca^{2+} channels themselves and/or vesicle priming due to remaining RIMs, Munc13, and other proteins, which can regulate recruitment of Ca^{2+} channels and synaptic vesicle priming without RIM-BP2. Instead, fine-tuning of release such as synchronization is compromised. We hypothesize that AZ scaffold might be differently built, leading to different consequences when eliminating one component. Hence deletion of RIM-BP2 leads to immediate loss of Ca^{2+} channels and/or vesicle priming. This is in line with the proposal that at tonic synapses, synaptic vesicles are only loosely coupled with Ca^{2+} channels (Vyleta and Jonas, 2014 [\[9\]](#)) and vesicles are de-primed under resting conditions (Miki et al., 2016 [\[12\]](#); Neher and Brose, 2018 [\[13\]](#)). Future research along this line may lead to an understanding of how fine molecular differences in the AZ scaffolds might orchestrate synaptic diversity.

Slice preparation

All animal experiments were conducted in accordance with the guidelines of the Physiological Society of Japan, and were approved by Doshisha University Animal Experiment Committee (A22063, D22063).

We used male and female C57BL/6 mice (postnatal days 35–40). WT and RIM-BP2 KO mice (Grauel et al., 2016 [DOI](#)) were deeply anesthetized with isoflurane and decapitated. Their brains were quickly removed and chilled in sherbet-like cutting solution containing (in mM): 87 NaCl, 75 sucrose, 25 NaHCO₃, 1.25 NaH₂PO₄, 2.5 KCl, 10 glucose, 0.5 CaCl₂ and 7 MgCl₂ bubbled with 95% O₂ and 5% CO₂ (Hallermann et al., 2003 [DOI](#)). Hippocampal slices (300 μm thick) were prepared from brains using a vibratome (VT1200S, Leica) in ice-cold cutting solution. After slicing, slices were incubated in cutting solution at 37°C for 30 min, and subsequently kept at room temperature (22–25°C) up to 4 h.

Whole-cell recordings

Electrophysiological recordings were performed in a recording chamber filled with the extracellular solution containing (in mM): 125 NaCl, 2.5 KCl, 25 glucose, 25 NaHCO₃, 1.25 NaH₂PO₄, 0.4 ascorbic acid, 3 myo-inositol, 2 Na-pyruvate, 2 CaCl₂ and 1 MgCl₂ saturated with 95% O₂ and 5% CO₂. In some experiments, the concentration of CaCl₂ was changed to 1 mM, 1.5 mM and 4 mM. For membrane capacitance measurements, 1 μM tetrodotoxin (TTX, Wako) was added to block Na⁺ channels. Moreover, for recording Ca²⁺ currents, 10 mM TEA-Cl was added to block K⁺ channels. Slices were visualized by an upright microscope (BX-51, Olympus). Whole-cell patch-clamp recordings were applied to hippocampal mossy fiber terminals at room temperature. The patch pipettes were filled with the intracellular solution containing (in mM): 135 Cs-gluconate, 20 TEA-Cl, 10 Hepes, 5 Na₂-phosphocreatine, 4 MgATP, 0.3 GTP and 0.5 EGTA (pH = 7.2 adjusted with CsOH). In some experiments, the concentration of EGTA was changed to 5 mM. Presynaptic patch pipettes (BF150-86-10, Sutter Instrument) had a resistance of 15–20 MΩ and series resistance (R_s) was 30–70 MΩ. R_s was compensated so that residual resistance was 30–35 MΩ. Patch-clamp recordings were performed using an EPC10/3 amplifier (HEKA) in voltage-clamp mode, controlled by Patchmaster software (HEKA). Membrane currents were low-pass filtered at 2.9 kHz and sampled at 20 kHz. Membrane capacitance measurements were performed using an EPC10/3 amplifier in the sine + DC configuration (Gillis, 2000) using Patchmaster software (HEKA). A sine wave (30 mV in amplitude, 1000 Hz in frequency) was superimposed on the holding potential of –80 mV. Data were obtained from at least three different WT and RIM-BP2 KO terminals in each set of experiments (biological replicate).

Immunostaining

Brains of WT and RIM-BP2 KO mice were quickly removed and transferred to cryomold (Sakura Finetek) filled with tissue freezing medium (Leica). Brains were instantaneously frozen on aluminum block in liquid nitrogen and stored at –80°C. For cryosectioning, frozen brains were kept in cryostat (CM1860, Leica) at –18°C for 30 min. Then, 10 μm-thick sections were sliced and collected on cover glasses (Matsunami glass, 25×25 No.1). Sections were quickly fixed by dehydration with a heat blower at 50°C for 1 min, and further dehydrated in ethanol for 30 min at –25°C and in acetone for 10 min on ice. After blocking with PBS containing 0.3% BSA at room temperature, sections were incubated with primary antibodies diluted in PBS containing 0.3% BSA: anti-VGLUT1 (1:2000, Synaptic Systems, guinea pig polyclonal, RRID:AB_887878), anti-PSD-95 (1:50, NeuroMab, Mouse monoclonal IgG2a, clone K28/43, TC Supernatant, RRID:AB_2877189), anti-Munc13-1 (1:1000, Mouse monoclonal IgG1, clone 11B-10G, Sakamoto et al., 2018 [DOI](#)), and anti-Cav2.1 (1:400, Synaptic Systems, rabbit polyclonal, RRID:AB_2619841) or anti-Cav2.2 (1:400, Synaptic Systems, Mouse monoclonal IgG2b, clone 163E3, RRID:AB_2619843) or anti-RIM-BP2

(1:200, Mouse monoclonal IgG2b, clone 8-4G, [Sakamoto et al., 2022](#)) for 3 hours at room temperature. Afterwards, sections were washed and incubated with fluorescence dye labeled secondary antibodies: DyLight 405 Anti-Guinea Pig IgG (Jackson ImmunoResearch, RRID:AB_2340470), Alexa Fluor 488 (Thermo Fisher Scientific) labeled Anti-Mouse IgG2a (Jackson ImmunoResearch, RRID:AB_2338462), STAR635P (Abberior) labeled Anti-Mouse IgG1 (Jackson ImmunoResearch, RRID:AB_2338461), and Alexa Fluor 594 (Thermo Fisher Scientific) labeled Anti-Rabbit IgG (Jackson ImmunoResearch, RRID:AB_2340585) or Anti-Mouse IgG2b (Jackson ImmunoResearch, RRID:AB_2338463) for 1 hour at room temperature. Sections were post-fixed with 4% PFA in PBS and washed with PBS. Sections were mounted on coverslips using Prolong Glass Antifade Mountant (Thermo Fisher Scientific). The same number of sections were obtained from WT and RIM-BP2 KO brains and processed in parallel at the same experimental day (biological replicate).

STED imaging

STED imaging was performed using TCS SP8 STED 3x microscope (Leica) equipped with a 405 nm diode laser, a pulsed white light laser (WLL), a continuous 592 nm STED laser for alignment, a pulsed 775 nm STED laser, HyD detectors, and a 100× oil-immersion objective lens (NA = 1.4). STED images were acquired using the Leica LAS-X software with an image format of 1024×1024 pixels, 16-bit sampling, 8-line accumulations, and 11.36-zoom factor, yielding a pixel size of 10 nm. HyD detectors were configured to counting mode with a gating from 0.5 to 6.5 nanosecond. A 405 nm diode laser was used to excite DyLight405. Alexa488, Alexa594, and STAR635P were excited using WLL at 488 nm, 561 nm, and 633 nm, respectively. The 775 nm STED laser power was set to 75% and 100% of maximum power for depletion of STAR635P and Alexa594, respectively, and delay time was set to 300 picoseconds. STED imaging was performed on thin sections obtained from four different WT and RIM-BP2 KO mice (biological replicate). Images were acquired three times from each section (technical replicate).

Analysis

For electrophysiological experiments, the data were analyzed by Igor Pro (WaveMetrics) or Excel (Microsoft Corp). To determine the time constant of release, an exponential fit was applied. Values are given as mean ± SEM, and n indicates the number of recorded terminals. Statistical method of sample size determination was not done, but our sample sizes are similar to those of previous studies ([Hallermann et al., 2003](#), [Midorikawa and Sakaba, 2017](#)).

For STED imaging, data were analyzed using custom-designed programs in Mathematica (Wolfram), values were processed by Igor Pro (WaveMetrics), and pictures were created with ImageJ (NIH). To determine the area of AZs, image masks were generated from STED images of Munc13-1 by unsharp masking and image binarization. Only AZs that show co-localization with VGLUT1 and PSD-95 immunofluorescence signals were included in the analysis. To confine the analysis to the large mossy fiber boutons, AZs in presynaptic terminals of small VGLUT1-positive area were excluded. Then, the background subtracted integral of signal intensity of Cav2.1 or Cav2.2 was quantified using the image masks of AZs. The background signal intensity was estimated from non-AZ area in the same STED images. For quantification of Cav2.1 clusters in the AZs, STED images of Cav2.1 were deconvolved using a gaussian kernel with a radius of 40 nm. Image masks of each Ca²⁺ channel were generated from deconvolved STED images by unsharp masking and image binarization. Then, the total signal intensity, the size, and the number of Cav2.1 clusters at the AZs were quantified using the image masks of Cav2.1 clusters. Custom-designed programs in Mathematica (Wolfram) are available from Source code 1. Values are given as mean ± SEM, and n indicates the number of animals. Statistical method of sample size determination was not done, but our sample sizes are similar to those of previous studies ([Brockmann et al., 2019](#)).

Experiments were not fully randomized and blinded.

Statistical analysis was done in MATLAB (The MathWorks). To consider the data variability among terminals, Wilcoxon rank sum test was used for significance test of two groups. Significance level was set at $\alpha = 0.05$, and p values were described in the main text or figure legends.

Data availability

All data generated in this study are included in the main text or shown as scatter plots in each graph. Numerical values of graphs are provided in Source data files. The custom-written code files in Mathematica are uploaded as Source code file.

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Competing interests

None

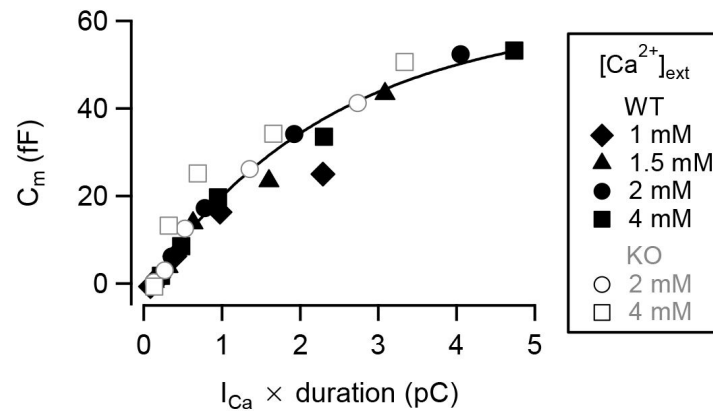


Figure 3-figure supplement 1.

The relationship between synaptic vesicle release and total Ca^{2+} charge

Capacitance jumps elicited by various pulse durations are plotted against total Ca^{2+} charge. The total Ca^{2+} charge is calculated by multiplying a peak Ca^{2+} current amplitude by pulse duration. Each symbol represents a data point obtained from WT (black filled symbols) and RIM-BP2 KO (gray open symbols) mice at various $[\text{Ca}^{2+}]_{\text{ext}}$. Data points were fitted by a single exponential function. Numerical values of plots are provided in [Figure 3-figure supplement 1-source data 1](#). [Figure supplement 1-source data 1](#). Datasets of Ca^{2+} charge and ΔC_m presented in [Figure 3-figure supplement 1](#).

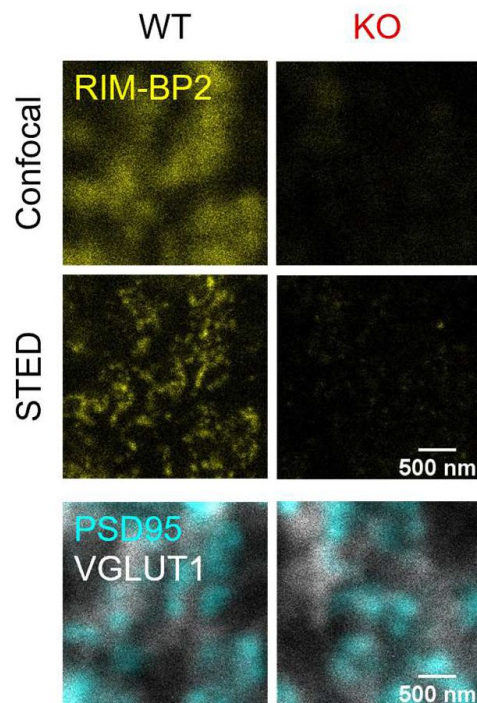


Figure 5-figure supplement 1.

STED microscopy confirmed loss of RIM-BP2 proteins in KO terminals

Confocal (*top*) and STED (*middle*) images of RIM-BP2 (yellow) in WT (*left*) and RIM-BP2 KO (*right*) hippocampal mossy fiber terminals identified by PSD95 (cyan) and VGLUT1 (white) (*bottom*). (Scale bar: 500 nm.) The *bottom* images show the same region as *top* and *middle*.

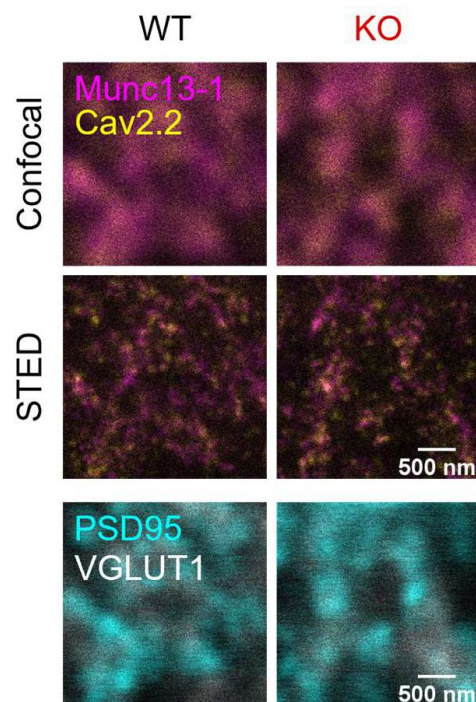


Figure 5-figure supplement 2.

STED imaging of Cav2.2 at hippocampal mossy fiber terminals

Confocal (*top*) and STED (*middle*) images of Munc13-1 (magenta) and Cav2.2 (yellow) clusters in WT (*left*) and RIM-BP2 KO (*right*) mice. Confocal images of PSD95 (cyan) and VGLUT1 (white) in the same region as *top* and *middle* are also shown (*bottom*). (Scale bar: 500 nm.)

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Article and author information

Rinako Miyano

Graduate School of Brain Science, Doshisha University, Kyotanabe-shi, Kyoto 610-0394, Japan
ORCID iD: [0000-0002-0022-7198](https://orcid.org/0000-0002-0022-7198)

Hirokazu Sakamoto

Department of Pharmacology, Graduate School of Medicine, The University of Tokyo, Bunkyo-ku, Tokyo 113-0033, Japan
ORCID iD: [0000-0002-9881-5173](https://orcid.org/0000-0002-9881-5173)

Kenzo Hirose

Department of Pharmacology, Graduate School of Medicine, The University of Tokyo, Bunkyo-ku, Tokyo 113-0033, Japan, International Research Center for Neurointelligence (WPI-IRCN), The University of Tokyo, Bunkyo-ku, Tokyo 113-0033, Japan
ORCID iD: [0000-0002-8944-6513](https://orcid.org/0000-0002-8944-6513)

Takeshi Sakaba

Graduate School of Brain Science, Doshisha University, Kyotanabe-shi, Kyoto 610-0394, Japan
For correspondence: tsakaba@mail.doshisha.ac.jp
ORCID iD: [0000-0003-0688-7717](https://orcid.org/0000-0003-0688-7717)

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Editors

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Reviewer #1 (Public Review):

This nice study by Miyano combines slice electrophysiology and superresolution microscopy to address the role of RBP2 in Ca²⁺ channel clustering and neurotransmitter release at hippocampal mossy fiber terminals. While a number of studies demonstrated a critical role for RBPs in clustering Ca²⁺ channels at other synapses and some provided evidence for a role of the protein in molecular coupling of Ca²⁺ channels and release sites, the present study targets another key synapse that is an important model for presynaptic studies and offers access to a microdomain controlled synaptic vesicle (SV) release mechanism with low initial release probability.

Summarizing a large body of high-quality work, the authors demonstrate reduced Ca²⁺ currents and a reduced release probability. They attribute the latter to the reduced Ca²⁺

influx and can restore release by increasing Ca^{2+} influx. Moreover, they propose an altered fusion competence of the SVs, which is not so strongly supported by the data in my view.

The effects are relatively small, but I think the careful analysis of the RBP role at the mossy fiber synapse is an important contribution.

Reviewer #2 (Public Review):

The proper expression and organization of CaV channels at the presynaptic release sites are subject to coordinative and redundant control of many active zone-specific molecules including RIM-BPs. Previous studies have demonstrated that ablation of RIM-BPs in various mammalian synapses causes significant impairment of synaptic transmission, either by reducing CaV expression or decoupling CaV from synaptic vesicles. The mechanisms remain unknown.

In the manuscript, Sakaba and colleagues aimed to examine the specific role of RIM-BP2 at the hippocampal mossy fiber-CA3 pyramidal cell synapse, which is well-characterized by low initial release probability and strong facilitation during repetitive stimulation. By directly recording Ca^{2+} currents and capacitance jumps from the MF boutons, which is very challenging but feasible, they showed that depolarization-evoked Ca^{2+} influx was reduced significantly (~39%) by KO of RIM-BP2, but no impacts on Ca-induced exocytosis and RRP (measured by capacitance change). They used STED microscopy to image the spatial distribution of the CaV2.1 cluster but found no change in the cluster number with a slight decrease in cluster intensity (~20%). They concluded that RIM-BP2 functions in tonic synapses by reducing CaV expression and thus differentially from phasic synapses by decoupling CaV-SV.

In general, they provide solid data showing that RIM-BP2 KO reduces Ca influx at MF-CA3 synapse, but the phenotype is not new as Moser and colleagues have also used presynaptic recording and capacitance measurement and shown that RIM-BP2 KO reduces Ca^{2+} influx at hair cell active zone (Krinner et al., 2017), although at different synapse model expressing CaV1.3 instead of CaV2.1. Further, the concept that RIM-BP2 plays diverse functions in transmitter release at different central synapses has also been proposed with solid evidence (Brockmann et al., 2019).